Jianqiao Lu

+86 15315884246 /+852 63149434 | tjlujianqiao@gmail.com | Website | Github

Summary

I am a Research Engineer at **ByteDance Seed Foundation Model**, working on **scaling and efficiency** for large-scale foundation models. My focus includes (1) training stability and architecture/system co-design for large-scale pretraining, (2) **long-context** modeling and efficient inference/prefill, and (3) post-training for capability and reliability (SFT/RL), with a growing interest in **agentic systems**. Previously, I conducted research on mathematical reasoning and formal verification with LLMs (NeurIPS/ICLR/ACL/EMNLP).

Education

- **The University of Hong Kong** *2020 - 2025*
  PhD, Computer Science
- **Tongji University** *2015 - 2020*
  Bachelor, Electronics

Experience

- **ByteDance** *2025 - Present*
  Research Engineer (TopSeed Program), Seed Foundation Model Beijing, China
  - Contributed to architecture & training design for large-scale foundation model pretraining.
  - Improved training stability and model quality versus prior baselines.
  - Worked on scaling & efficiency: optimizing throughput/cost, and improving long-context training/inference pipeline.
- **ByteDance** *04/2024 - 2025*
  Research Intern, Seed Foundation Model Beijing, China
- **Huawei** *04/2022 - 04/2024*
  Research Intern, Noah's Ark Lab Shenzhen, China

Research

Long Context Prefilling

- ***(Equal Contribution)*** *FlexPrefill: A Context-Aware Sparse Attention Mechanism for Efficient Long-Sequence Inference*
  ICLR 2025 Oral
  We propose context-aware sparse attention that adapts its patterns and computational budget to each input and attention head, reducing prefill cost for long-sequence inference while balancing latency and model quality across the evaluated models and tasks.

AI for Math

- ***(First Author)*** *AutoPSV: Automated Process-Supervised Verifier*
  NeurIPS 2024
  An automated process labeling system that significantly enhances the accuracy of reasoning models by detecting and leveraging confidence shifts in reasoning steps, resulting in improvements of up to 34% over self-consistency across math and commonsense benchmarks.

- ***(Equal Contribution)*** *FormalAlign: Automated Alignment Evaluation for Autoformalization*
  ICLR 2025
  A framework that automates the evaluation of semantic alignment between natural and formal languages in autoformalization, outperforming GPT-4 by 11.58% on FormL4-Basic and 3.19% on MiniF2F-Valid, significantly reducing the reliance on manual verification.

- *Proving Theorems Recursively*
  NeurIPS 2024
  Developed POETRY, a recursive proof method that boosts success rates by 5.1% and doubles proof length on miniF2F.

- *FVEL: Interactive Formal Verification Environment with Large Language Models via Theorem Proving*
  NeurIPS 2024 (Datasets and Benchmarks Track)
  Introduced FVEL, an interactive formal verification environment that integrates LLMs with Isabelle for neural automated theorem proving, resulting in a 17.39% improvement in problem-solving on SV-COMP and a reduction in proof errors.

Multi-Modality (Speech & Text)

- ***(First Author)*** *Improving End-to-End Speech Processing by Efficient Text Data Utilization with Latent Synthesis*
  EMNLP 2023
  Developed the Latent Synthesis framework to efficiently utilize textual data for enhancing end-to-end speech processing models, achieving over 22.3% reduction in word error rate for ASR and significant improvements in SLU tasks.

Benchmarking

- *MR-Ben: A Meta-Reasoning Benchmark for Evaluating System-2 Thinking in LLMs*
  NeurIPS 2024
  MR-Ben benchmark consists of 5,975 expert-curated questions across multiple domains to evaluate the meta-reasoning capabilities of LLMs.

- *Planning, Creation, Usage: Benchmarking LLMs for Comprehensive Tool Utilization in Real-World Complex Scenarios*
  ACL 2024
  A benchmark that evaluates LLMs' abilities in planning, creating, and using tools within real-world scenarios. UltraTool emphasizes complex, multi-step tasks, offering a more realistic assessment of LLMs' tool utilization capabilities beyond simple, synthesized queries.
  Code

Online Matching

- ***(Equal Contribution)*** *Online Matching Meets Sampling Without Replacement*
  WINE 2024
  This work provides the first competitive analysis of this method, showing its effectiveness in both Online Bipartite Matching and Online Stochastic Matching problems.